

ANWAR HUSSAIN SOFI

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Aim

AI researcher specializing in trustworthy computer vision for clinical decision support, with a focus on interpretable medical image analysis. I bring research experience in medical imaging, vision-language models, and clinically oriented AI evaluation. I am seeking a First Stage Researcher/PhD position to further develop my expertise on interpretable, reliable, and clinically translatable AI systems.

Research Interests

Deep Learning • Computer Vision • Medical Image Analysis • Machine Learning • Object Detection • Vision-Language Models • Neural Network Architecture • Biomedical AI Applications • AIOT

Education

National Taiwan Ocean University (NTOU), Keelung, Taiwan

Master of Science, Applied Artificial Intelligence

Expected 2026

Grade: A++

Thesis: *Assessment of Cleavage-Stage Embryo Development Using Ensemble Heterogeneous Machine Learning Models*

Advisor: Prof. Jung-Hua Wang,

- **EMBRACE-MTL**, a multi-task deep learning framework for joint cytoplasmic-fragmentation segmentation, t2/t4 developmental-stage classification, and three-class blastomere-symmetry grading.
- **SiameseCV**, a cross-view consistency learning framework that generates pixel-level fragmentation masks, estimates continuous fragmentation ratios, and assigns corresponding fragmentation grades.
- **Methods** ResNet-50 architecture with concatenation-based multi-scale feature fusion, a U-Net-style segmentation decoder, and task-specific classification heads for local and global embryo-morphology assessment.
- **Award:** NTOU International Master Program Scholarship (2024–2026).

Dr. M.G.R Educational and Research Institute, Chennai, India

Bachelor of Technology, Information Technology

2019 – 2023

Grade: A++ Achievement: Prime Minister's Special Scholarship (PMSSS) Holder

- **Key Courses:** Discrete Mathematics, Data Structures, Algorithms, Computer Architecture, Operating Systems, DBMS, Artificial Intelligence, Machine Learning, Cloud Computing.
- **Award:** Sole recipient from hometown of fully-funded merit scholarship by AICTE, Govt. of India.

Research Experience & Projects

Research Highlights (New Developments)

- **Automated Multi-Feature Description Generation for Cleavage-Stage Embryo Fragmentation Analysis Using Vision-Language Models and Deep Segmentation** *2026*

Automated embryo fragmentation analysis combining deep segmentation (AttnRegDeepLab) with vision-language models (Moondream2), generating multi-feature clinical descriptions from microscopy images. Achieved **0.70+ Dice score** with explainable AI framework. • **Comprehensive Deep Learning-based Approach for Quality Assessment of Blastocyst** 2025

Developed an AI tool that helps IVF doctors assess embryo quality from time-lapse images. The system automatically tracks embryo growth stages and measures key features that indicate pregnancy success, achieving **39% better accuracy** than existing methods and reducing subjective human error in clinical decisions. • **Assessment of Cleavage-Stage Embryo Development Using Ensemble Hetero Machine Learning Models** 2024

Developed an ensemble AI platform that automatically tracks early embryo development stages in IVF. Used **YOLOv12** for cell detection and **Inception ResNet V2** for stage classification, improving accuracy in identifying cell division timing and reducing observer bias in embryo assessment.

Other Key Projects

AI Assistant for Automatic Interpretation of Reproductive Medical Images 2025
Excellent Award Winner, Best AI Awards International Competition

- Developed a vision-language model for medical imaging interpretation, integrating multimodal AI architectures to automate the analysis of reproductive health data.

ST-MIND: AI-Based Interactive Student Educational Application 2023
Published in Journal of Interdisciplinary Cycle Research

- Architected an AI-driven platform using Python and Django to handle academic question resolution and facilitate collaborative learning between students and faculty.

Fake News Detection Framework

- Built a machine learning framework to identify counterfeit news by analyzing linguistic patterns, source credibility, and social network dynamics using NLP techniques.

Professional Experience & Internships

Data Science Intern April 2025 – Aug 2025
Codtech IT Solutions Pvt. Ltd.

- Developed and deployed machine learning models, mastering Python and SQL for data-driven business solutions.
- Applied advanced data preprocessing and optimization pipelines to solve real-world problems.

Machine Learning Intern Feb 2023 – June 2023
L&T EduTech, Chennai, India

- Focused on Neural Networks and Deep Learning classifiers, building foundational skills in algorithm development and model training.
- Gained practical exposure to real-time model evaluation and dataset management.

2025 NCHU Semiconductor Summer School

National Chung Hsing University (NCHU), Taiwan Aug. 11 – Aug. 22, 2025

- **Specialized Training:** Completed an intensive 10-day program covering IC Design, Semiconductor Manufacturing, Materials, and Packaging & Testing.
- **Hands-on Labs:** Gained practical experience with Photomask Design, Scanning Electron Microscope (SEM), and Atomic Force Microscope (AFM) operations.

- **Advanced Topics:** Attended technical lectures on Green Technology, Solar Cell Semiconductor Devices, and Future Semiconductor Technologies.

Web Developer & Digital Marketing Intern

Mar 2022 – June 2022

Hypstuma

- Designed and hosted responsive websites; managed end-to-end deployment and digital marketing.

Technical Skills

- **Programming Languages:** Python (Proficient), C, C++, Java, SQL, JavaScript.
- **Deep Learning:** TensorFlow, PyTorch, Keras, YOLOv12, SAHI, Inception-ResNet-V2.
- **Data Analytics:** Pandas, NumPy, Tableau, Power BI, Excel (VBA).
- **Web Development:** HTML, CSS, Django, Flask, Bootstrap.
- **Tools:** Git/GitHub, VS Code, PyCharm, Anaconda, Jupyter Notebook, Google Colab.
- **Operating Systems:** Windows, Linux, macOS.

Awards & Achievements

- **Excellent Award (2025):** Best AI Awards International Competition.
- **NTOU Scholarship Holder (2024-2026):** Academic excellence in Master's program.
- **PMSSS Scholarship Holder (2019-2023):** Fully-funded merit scholarship by AICTE, Govt. of India.
- **Certifications:** Python Pro Bootcamp (Udemy), Data Science Specialization (Great Learning).

References

Prof. Jung-Hua Wang

Full Professor

Department of Electrical Engineering

National Taiwan Ocean University

Expertise: Artificial Intelligence, Machine Learning, Deep Learning, and Computer Vision

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Prof. Chin-Chun Chang

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